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I. Introduction

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II. Literature Review

A. Evolution of Speech Representation Learning

Speech representation learning has evolved from fixed acoustic features to models pretrained on unlabeled audio [1]–[3]. Self-supervised methods can also be organized according to the targets that they learn to predict [3]–[7].

1) Handcrafted and Statistical Models: Early speech-recognition systems divided audio into short frames and used fixed mathematical transformations to extract salient features [1], [8]–[10]. Mel-frequency cepstral coefficients (MFCCs) compressed the signal according to properties of human hearing, producing compact feature representations [8]. Perceptual linear prediction (PLP) incorporated further auditory characteristics, improving robustness across speakers and recording conditions [9]. Researchers also captured temporal changes in speech by computing differences between adjacent features [8], [9]. These features were passed to statistical models: hidden Markov

models (HMMs) represented speech timing and sequence [10], whereas Gaussian mixture models estimated acoustic likelihoods [1]. Although computationally efficient, this pipeline separated feature extraction, acoustic modeling, and language modeling, preventing joint optimization [1], [10].

2) Supervised Methods: Supervised methods learn directly from transcribed speech [2], [11]–[15]. Deep neural networks replaced many older statistical components and improved recognition accuracy [2]. Convolutional neural networks captured local acoustic patterns [11], while recurrent neural networks modeled temporal context [12]. Connectionist temporal classification (CTC) enabled training without exact frame-level alignments [13]. Listen, Attend and Spell mapped audio directly to text using attention [14], and the Conformer combined self-attention with convolution to model both global context and local detail [15]. Despite simplifying the recognition pipeline and improving accuracy, these methods still required substantial quantities of labeled audio [2], [14], [15].

3) Self-Supervised Methods: Self-supervised methods learn from raw audio without transcripts by deriving training objectives from the signal itself [3]–[7], [16]–[18]. Contrastive predictive coding (CPC) selected the correct future representation from negative candidates [16], whereas autoregressive predictive coding (APC) predicted future acoustic frames directly [17]. Mockingjay [18] and TERA [4] masked or altered portions of the input and learned to reconstruct them. Later, wav2vec 2.0 applied contrastive prediction to quantized speech representations [3]; HuBERT predicted cluster assignments for masked regions [5]; and data2vec predicted teacher-network representations [7]. These models can pretrain on unlabeled audio and then adapt to downstream tasks with substantially less labeled data [3], [5].

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B. Self-Supervised Learning Paradigms

Self-supervised speech-learning methods can be grouped into four categories according to their prediction targets [3]–[7].

1) Reconstruction: Reconstruction methods mask or corrupt parts of an audio input and train the model to recover the missing information [4], [18], [19]. Mockingjay masks acoustic frames and uses a Transformer encoder to reconstruct them [18]. DeCoAR uses bidirectional context to reconstruct acoustic frames [19], while TERA applies alterations along the time, frequency, and magnitude dimensions to improve robustness [4]. This objective is straightforward and requires neither negative sampling nor clustering [4], [18]. However, the model may devote capacity to background noise and low-level acoustic details that are less relevant to speech recognition [4], [19].

2) Contrastive Learning: Contrastive learning trains a model to distinguish the correct latent target from negative candidates [3], [16]. CPC uses the current context to identify the true future representation among distractors [16]. Wav2vec 2.0 masks portions of the latent speech sequence and selects the corresponding quantized representation from distractors sampled from the same utterance [3]. Predicting high-level latent representations can be more useful than reconstructing raw acoustic input [3], [16]. However, performance depends on negative-sampling choices, and the quantization codebook adds training complexity [3].

3) Discrete Unit Prediction: Discrete-unit prediction converts continuous speech into categorical targets and trains the model to predict those targets [5], [6]. HuBERT clusters acoustic features, masks portions of the input, and predicts cluster assignments for the masked regions [5]. WavLM builds on masked speech prediction and augments training with noise and overlapping speech to improve performance under challenging acoustic conditions [6]. Categorical prediction aligns well with speech recognition and avoids explicit negative sampling [5], [6]. Its quality nevertheless depends on the initial clustering or pseudo-labeling process; noisy targets can limit the learned representations [5], [6].

4) Representation Prediction: Representation-prediction methods train a model to predict continuous feature vectors rather than reconstructing raw audio or predicting discrete labels [7], [20]–[22]. In data2vec, a student network predicts contextualized representations produced by a teacher network that observes the complete input [7]. Joint-embedding predictive architectures (JEPAs) [20], [21] and Audio-JEPA [22] use a context encoder to predict target-network representations of masked input regions. This approach avoids reconstructing low-level noise, selecting negative examples, and relying on externally generated cluster assignments, while retaining information useful for downstream tasks [3], [5], [22].

C. Maintaining the Integrity of the Specifications

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Acknowledgment

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